

Trivia Quiz System - Full Code Documentation

This document explains the complete structure and functionality of the Trivia Quiz System project. The project was developed in C++ using Object-Oriented Programming concepts such as classes, inheritance, templates, STL containers, operator overloading, and file handling.

1. Project Structure

The project contains two applications:

main.cpp – Player application

Handles quiz gameplay, answering questions, saving progress, and leaderboard viewing.

admin.cpp – Admin application

Handles question management, leaderboard management, and progress reset.

2. Libraries Used

iostream

Used for user input and output with cout and cin.

fstream

Used for reading and writing files using ifstream and ofstream.

vector

STL dynamic container used to store questions and players.

string

Used for text handling.

algorithm

Used for sorting leaderboard data.

3. Question Class

The Question class stores all information related to a quiz question.

Data Members:

- string text
- string options[4]
- int correctAnswer
- int difficulty

The class also contains constructors and operator overloading.

4. Player Class

The Player class stores player information.

Data Members:

- string name
- int score

Main Functions:

- incrementScore()
- saveToLeaderboard()

5. GamePlayer Class

GamePlayer inherits from Player using inheritance.

Additional Features:

- saveProgress()
- loadProgress()
- currentQuestion tracking

Inheritance allows GamePlayer to reuse Player data and functions.

6. Admin Class

The Admin class is responsible for administrative functionality.

Main Features:

- Add questions
- Delete questions
- View leaderboard
- Reset progress

The admin application also contains password authentication.

7. File Handling

The project uses text files to permanently store data.

questions.txt

Stores all quiz questions.

leaderboard.txt

Stores player names and scores.

progress.txt

Stores saved quiz progress.

File handling is implemented using:

- ifstream
- ofstream

8. STL Vector

STL vectors are used to dynamically store objects.

Example:

```
vector questions;
```

Vectors automatically resize during runtime and are part of the Standard Template Library.

9. Operator Overloading

Operator overloading is implemented using:

```
friend ostream& operator<<(ostream& out, const Question& q)
```

This allows objects to be displayed directly using:

```
cout << question;
```

10. Template Function

Template functions are used to reuse code for multiple data types.

Example:

```
template void displayList(...)
```

This function can display lists of Questions, Players, and other objects.

11. Leaderboard System

The leaderboard stores player scores.

Scores are sorted using:
`sort()`

The system displays players in descending order based on score.

12. Main Game Logic

The main game loop:

1. Loads questions from file
2. Displays questions
3. Receives player answers
4. Checks correctness
5. Updates score
6. Saves progress
7. Updates leaderboard

13. OOP Concepts Used

The project uses:

- Classes and Objects
- Inheritance
- Encapsulation
- Templates
- Operator Overloading
- STL Containers
- File Handling

14. Conclusion

The Trivia Quiz System demonstrates a complete C++ application using Object-Oriented Programming principles. The project successfully combines gameplay, administration, persistent storage, reusable code, and structured design.